

Haowen Yan

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BIOGRAPHY

I am Haowen Yan, a third-year student majoring in Computer Science and Technology at Sun Yat-Sen University. Currently, I am a research intern at Xspark AI. I am also privileged to intern at TEA Lab under the supervision of Prof. Huazhe Xu, where I worked on teleoperation. Before that, I was also fortunate to be mentored by Prof. Gang Chen, working on optimal control and UAV.

My research interests focus on **Embodied AI**, **Robotics**, **Reinforcement Learning**, **World Models**, and **Optimal Control**.

EDUCATION

Sun Yat-Sen University

Bachelor in Computer Science

Guangdong, China

Sept 2023 – Present

- **GPA:** 4.3397/5.0; **Rank:** 3/353 (top 0.85%)

- **Course Grades:**

Control Theory (100/100), Optimization Theory (99/100), Mathematics Analysis II (98/100), Machine Learning (98/100), Data Structures (98/100), Signals and Systems (97/100), Probability and Statistics (96/100), etc.

- **Self-Study Courses based on Interest:**

Reinforcement Learning, Optimal Control, Robotics, etc.

PUBLICATIONS

* indicates equal contribution.

- Tianxing Chen*, Zixuan Li*, Weijie Wan*, **Haowen Yan***, et al. "RoboDojo: A Unified Sim-and-Real Benchmark for Efficient Evaluation of Generalist Robot Manipulation Policies", **ICRA** Workshop, 2026.

AWARDS & HONORS

Honors

- **2024 National Scholarship** (Ranked 1st out of 159 students in comprehensive evaluation)
- **2025 National Scholarship** (Ranked 8th out of 352 students in comprehensive evaluation)
- **2024 First-Class Academic Scholarship** (Awarded to top 8% students in comprehensive evaluation)
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Competition Awards

• China Undergraduate Operating System Design Competition	National Third Prize	2025.08
• China Undergraduate Mathematics Competition	Provincial First Prize	2024.11
• Shenzhen Cup Mathematical Modeling Competition	Provincial Second Prize	2024.12
• China Undergraduate Mathematical Modeling Competition	Provincial Third Prize	2024.10
• Mathematical Contest in Modeling, USA	H Award	2025.05
• MathorCup Mathematical Modeling Competition	Provincial Second Prize	2024.04

RESEARCH EXPERIENCE

RoboDojo: A Unified Sim-and-Real Benchmark for Efficient Evaluation of Generalist Robot Manipulation Policies
2026.03–2026.05

Advisors: Prof. Wenbo Ding

Tsinghua University

- **Introduction:** To improve the standardization and fairness of robot evaluation, we present RoboDojo, a unified sim-and-real benchmark for evaluating generalist robot manipulation policies.
- **Involvement:**
 - Designed a standardized real-robot evaluation platform, ensuring consistency and reproducibility of the evaluation environment across all aspects and details to facilitate large-scale benchmarking.
 - Trained mainstream policies (including π_0 , $\pi_{0.5}$, ACT, etc.) and evaluated them on real robots in the RoboDojo platform.
 - Designed a web dashboard for live monitoring of evaluation progress and for viewing participants' evaluation video replays and related metadata.
 - Implemented an object reset routine on the real-robot evaluation platform to ensure items are placed at the same locations before each trial, guaranteeing fairness and consistency of evaluations.
- **Outcomes:**
 - The paper is currently under review for publication at ICRA 2026.
 - The evaluation platform is currently applying for a national utility-model patent.

Ultra-Fast Model Predictive Controller (MPC)

2024.12–2025.12

Advisors: Prof. Gang Chen

Sun Yat-sen University

- **Introduction:** Model Predictive Control (MPC) is a robust control method, but its high computational cost limits real-time deployment. We propose a hardware–software co-design approach to accelerate MPC and validate it through UAV experiments.
- **Involvement:**
 - Modeled UAV dynamics and verified state-space equations via MATLAB simulations.
 - Programmed the MPC on an FPGA and controlled UAV flight through ROS + Gazebo simulation.
 - Deployed a hardware–software co-accelerated MPC on a UAV and tested its performance.
- **Outcomes:**
 - This project awarded a provincial innovation grant.
 - Related paper is being prepared for submission.

Operating System Based on the RISC-V Instruction Set

2025.04–2025.08

Advisors: Prof. Pengfei Chen

Sun Yat-sen University

- **Introduction:** Developed an OS on RISC-V, supporting up to 90 system calls; participated in the Operating System Design Competition.
- **Involvement:**
 - Participated in the overall architecture design of the operating system.
 - Implemented most process management related system calls.
 - Adapted and debugged low-level drivers.
- **Outcomes:**
 - Project won the National Third Prize in the 2025 China Undergraduate Operating System Design Competition.

Research on Teleoperation of A Robotic Arm

2026.01–2026.02

Advisors: Prof. Huazhe Xu

Tsinghua University

- **Introduction:** In many existing teleoperation systems, the robot motion is affected by the head-mounted display (HMD): even when the controller remains stationary, movement of the HMD alone can cause unintended robot motion.
- **Involvement:**
 - Implemented teleoperation of an existing robotic arm using Meta Quest 3S based on an incremental control scheme.

- By introducing the initial pose of the HMD as a reference frame, this work eliminates the influence of HMD motion, thereby improving system stability and control accuracy.

Research on Diffusion Models

2025.12–2026.01

Advisors: Prof. Qinliang Su

Sun Yat-sen University

- **Introduction:** Studied diffusion models and implemented DDPM, DDIM, and DPM-Solver (1st-, 2nd-, and 3rd-order) samplers, and improved sampling efficiency.
- **Involvement:**
 - Used Pytorch to implement DDPM, DDIM and DPM-Solver (1st/2nd/3rd-order) samplers.
 - Trained and tested them on the Smithsonian Butterflies subset dataset.
 - Improved DPM-Solver with v-prediction to accelerate sampling and enhance sample quality.

Research on Single-cycle CPU and 5-stage pipelined CPU (MIPS)

2024.12–2025.01

Advisors: Prof. Zhi Zhou

Sun Yat-sen University

- **Introduction:** Designed and implemented a single-cycle CPU and a 5-stage pipelined CPU based on the MIPS architecture, covering the complete workflow from hardware design and verification to FPGA deployment.
- **Involvement:**
 - Designed a single-cycle CPU and a 5-stage pipelined CPU, with all hazards (data hazard, structural hazard, and control hazard) resolved in hardware.
 - Implemented both CPUs in Verilog, simulated and debugged them
 - Deployed CPUs on an FPGA development board, respectively.

SKILLS

English:	CET-4 (622), CET-6 (556)
Programming:	Python, Pytorch, C/C++, Matlab, Rust, Verilog
Software:	Linux, Mujoco, ROS, Gazebo, Git
Traits:	Innovative; diligent and detail-oriented; self-learner; persistent and hardworking